
Air Quality Prediction in Moroccan Cities

Project Goal: Build machine learning models to predict **air quality (PM2.5 or AQI)** in major Moroccan cities using real-world, publicly available data collected via APIs. Throughout this project, you will collect real data, build predictive models in a **centralized setting**, and then (optionally) **experimentally explore** the basics of **federated learning**.

Recommended Tools:

- Python (Jupyter Notebooks)
- Libraries: pandas, scikit-learn, matplotlib, requests
- APIs: OpenWeatherMap (<https://openweathermap.org/api>)

Deliverables:

- Documented and reproducible code (Jupyter Notebook or Python scripts)
- Clear and detailed explanations of each stage of the project

Project Instructions:

Step 1: Collect the Data

- Choose at least 3 Moroccan cities (e.g., Casablanca, Rabat, Marrakech, Fes).
- Use the OpenWeatherMap API to collect:
 - Weather data: temperature, humidity, pressure, wind speed, etc.
 - Air quality data: Air Quality Index (AQI), PM2.5
- Collect data for multiple timestamps (e.g., every hour or every few hours over several days).
- Store the data in structured format (e.g., CSV or database).

Step 2: Prepare the data

- Merge the collected data (weather data and air quality data) by city and timestamp
- Preprocess the dataset:
 - Remove or handle missing and duplicate records.
 - Convert data types and format columns as needed.
 - Encode categorical features (e.g., city name, weather description).
 - Normalize or scale numerical features if needed.
- Target variable: Choose either PM2.5 (regression) or AQI class (classification) as your prediction target.
- Prepare a clean dataset of input features and the target variable.

Step 3: Build the model

- Split your dataset into training and testing sets (e.g., 80/20).
- Train at least two different machine learning models (e.g., Linear Regression, Random Forest, Decision Tree, or a simple Neural Network).
- Evaluate model performance using appropriate metrics:
 - **Regression** (for PM2.5): MAE, RMSE, R^2
 - **Classification** (for AQI levels): Accuracy, F1-score, Confusion Matrix

Optional Bonus: Simulate federated learning

- Partition the dataset by city: each city acts as a federated client.
- Initialize a global model (e.g., LinearRegression).
- For each federated round:

1. Send the global model to each city (client).
 2. Each city:
 - Trains a local model on its city-specific data.
 - Returns learned parameters (coefficients and intercept).
 3. Aggregate the parameters using Federated Averaging (FedAvg):
 - Weighted average based on number of samples per city.
 - Update the global model parameters.
- After a few rounds, evaluate the final global model on test data.
 - Compare with the centralized model.